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REVIEW PAPER

Integrons: the hidden architects of bacterial adaptation, evolution, and the challenges of antimicrobial resistance

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Abstract Integrons, a diverse group of genetic elements, have emerged as key players in bacterial adaptation and evolution. These elements, commonly found in both environmental as well as clinical settings, facilitate the acquisition, exchange, and expression of integron cassettes, allowing bacteria to rapidly adapt to changing environments and acquire antibiotic resistance. This review provides an in-depth exploration of the various classes of clinical integrons, including class 1, 2, and 3, highlighting their origins, distribution, and associated mobile elements. We delve into the astonishing success of "class 1 integrons", emphasizing their ability to recognize diverse attachment sites known as "*attC* sites" and getting integrated within many different integron cassettes from diverse sources. Class 1 integrons are able to propagate widely among bacterial hosts due to their lack of host specificity, interaction with transposons, and broad host range plasmids. Moreover, we discuss the substantial impact of class 1 integrons in antimicrobial resistance, as they accumulate an array of resistance genes through strong positive selection. Additionally, we address the challenging issue regarding the evolution and function of integrons and integron cassettes, including the role of promoters, origins of integron cassettes, and the abundance of unknown proteins encoded within them. The future prospects of integron research are also explored, highlighting the need to understand cassette expression patterns, assess the contribution of chromosomal/superintegron arrays to host fitness, unravel the mechanisms of cassette generation, and investigate

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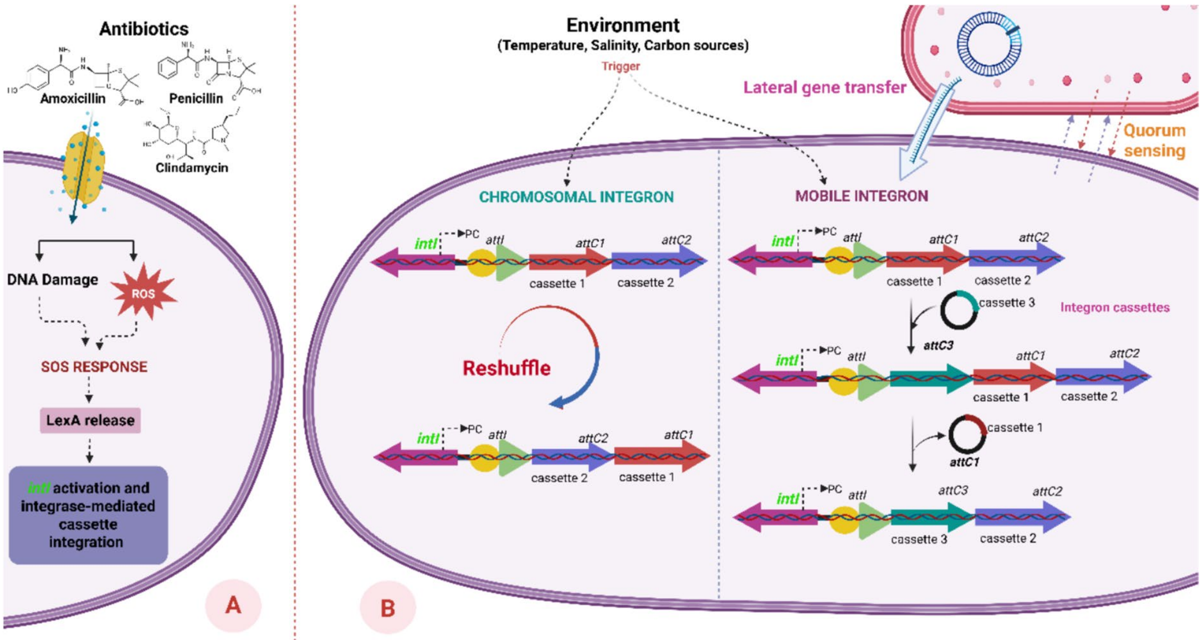
the connection between the SOS induction and horizontal gene transfer. Overall, this review underlines the significance of integrons as hidden architects driving bacterial adaptation and evolution, providing valuable insights into their ecological and evolutionary dynamics, and shaping the future direction of research in this field.

Graphical abstract **A** Antibiotics-mediated cascade activation; **B** Reshuffling of integron cassettes in sedentary chromosomal integrons and mobile integron cassettes transfer between bacteria for antimicrobial resistance



Keywords Antibiotic resistance · Bacterial evolution · Integron cassettes · Integrons · Antimicrobial resistance

Introduction

Over both short- and long-term evolutionary timescales, horizontal gene transfer is essential for modifying genomes (Wiedenbeck and Cohan 2011). Non-homologous recombination mechanisms have been found to be critical for adaptability in hostile settings, particularly in bacterial species. The remarkable ability of bacteria to acquire multiple genes conferring antibiotic resistance within relatively short periods is a striking example of this (Acar and Rostel 2001). Transposons, which facilitate the transfer of resistance genes between unrelated replicons, have been extensively investigated for their role in mobilizing and disseminating resistance to various antibiotics across diverse bacterial populations (Johansson et al. 2020). While transposition itself involves recombination, it often exhibits varying degrees of site specificity depending on the type of transposase involved. In comparison, integron-mediated site-specific recombination typically involves highly conserved target sequences, such as *attI* and *attC*, and recombination occurs between defined DNA loci (Hallet and Sherratt 1997). An exception to this model is the *integron*, a unique system wherein one of the two recombination sites belongs to a family of diverse attachment sites, providing access to a vast pool of adaptive genes that can be mobilized, rearranged, and disseminated among bacteria (Gillings et al. 2005). The *integron* potentially encompasses thousands of adaptive genes, offering bacterial populations a level of genome plasticity that other systems do not (Escudero José et al. 2015).

Integrons are broadly classified into two types: (i) Mobile integrons (MIs), which are capable of horizontal transfer between bacterial strains, and (ii) Sedentary chromosomal integrons (SCIs), which are integrated into the bacterial chromosome and are not transferable between strains. A well-characterized example of an SCI is found in the chromosome of *Vibrio cholerae*. Initially, five classes of mobile integrons were described, among which class 1, class 2, and class 3 integrons are most commonly associated with antibiotic-resistant clinical isolates (Hall 2001). The integron cassettes within these elements are discrete DNA segments that often carry functional genes. These cassettes can be captured, rearranged, and expressed by the *integron* system, frequently encoding antibiotic resistance determinants that enhance bacterial survival under selective pressures (Hall and Collis 1995; Joss et al. 2009).

Class 1 integrons, discovered in Gram-negative pathogens, exhibit a unique ability to capture and express external genes through site-specific recombination and internal promoters. These clinical integrons have been extensively characterized. However, it is important to emphasize that clinical class 1 integrons are not representative of integrons as a whole (Domingues et al. 2012; Nandi et al. 2004). Most integrons are located within bacterial chromosomes and contain integron cassettes encoding functions that are largely unknown. These integrons also exhibit conserved *attC* sites, suggesting a consistent structural motif across different systems (Rowe-Magnus et al. 2001). Moreover, SCIs may contain extensive cassette arrays with multiple promoters, in contrast to the shorter arrays commonly observed in clinical class 1 integrons (Fluit and Schmitz 2004).

Although much of our current understanding of integrons is derived from studies on clinical class 1 integrons, it is essential to expand research beyond these systems. Integrons are widespread across bacterial genomes, occupying diverse ecological niches and capable of horizontal transfer between species and lineages over evolutionary timeframes (Collis et al. 1993). They facilitate the acquisition and reassembly of diverse genes, including those conferring antibiotic resistance. This adaptability enables bacteria to respond rapidly to environmental pressures by incorporating beneficial genes into their genomes (Engelstädter et al. 2016). Notably, nearly two-thirds of the genes associated with integrons remain functionally uncharacterized (Buongermino Pereira et al. 2020). Therefore, it is of utmost importance to investigate the evolutionary trajectory of integrons, their behavior in the natural environments, and mechanisms governing integron cassette sampling and rearrangement. This will deepen our understanding of their biology, ecology, and evolution, as well as their potential adaptation strategies. By thoroughly examining the diversity and dynamics of integrons, we can better predict how these systems may evolve and be harnessed in future biotechnological applications.

Fundamental structure of integrons

Integrons represent a common genetic framework found in various bacteria, characterized by their ability to capture and express external genes through the integration of integron cassettes. Integrons are composed of two main components: (i) a functional platform and (ii) a variable array of integron cassettes. The functional platform, located at the 5' end, includes the integrase gene (*intI*), its promoter (PintI), an *attI* recombination site, and a constitutive cassette promoter (Pc) responsible for the expression of integron cassettes (Fig. 1). The *intI* gene encodes the integrase protein IntI, a site-specific tyrosine recombinase that catalyzes both the insertion and excision of integron cassettes by means of recombination, primarily occurring at *attI* and *attC* sites. Notably, *intI* is transcribed in the opposite direction to the integron cassettes within the integrin (Escudero José et al. 2015). However, in certain cases, integrons exhibit an unconventional arrangement in which the *intI* gene is transcribed in the same direction as the integron cassettes. These atypical systems, known as reverse or unusual integrons, have been observed in various bacterial species such as *Treponema denticola*, *Acinetobacter baumannii*, *Chlorobium phaeobacteroides*, and *Blastopirellula marina*. This unique configuration expands the diversity of integrons and highlights the adaptability of these genetic elements across different organisms (Boucher et al. 2007; Wu et al. 2013).

The other component of an integron is the integron cassette array. Each integron cassette typically contains a single open reading frame (ORF) along with

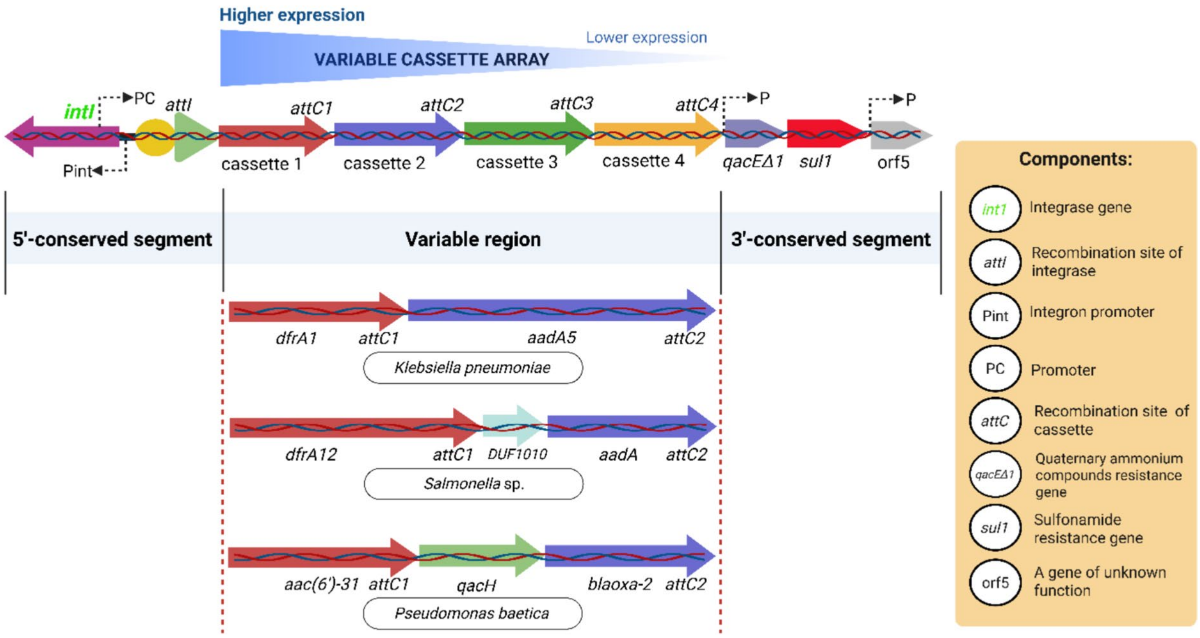


Fig. 1 The schematic depiction of the fundamental machinery of the "class 1 integrons", along with its functional components accountable for integration events;, functional diversity of "variable regions" of diverse microbial species highlighting the upshot of integrons for developing AMR (Kwiecień et al. 2020; Li et al. 2013; Possebon et al. 2021; Vásquez-Ponce et al. 2022)

a recombination site known as *attC*. Upon excision from the integron, an integron cassette transforms into a non-replicative mobile genetic element. This element serves as a substrate for recombination facilitated by the IntI enzyme, occurring between the *attI* site on the integron and the *attC* site on the circular integron cassette. This recombination process enables integration of the integron cassette into another integron, contributing to the diversification and dissemination of genetic material. Once successfully integrated, these integron cassettes can be expressed via the Pc promoter. This promoter initiates transcription of the integrated integron cassette, enabling the expression of its genetic content. Importantly, the integration process does not disrupt existing genes within the bacterial genome. Furthermore, once a gene cassette is integrated, it becomes part of the bacterial genome and is exposed to natural selection, which may favor traits that enhance bacterial adaptation (Collis and Hall 1992; Mazel 2006; Partridge et al. 2000; Tansirichaiya et al. 2019).

The integron system offers a remarkable capacity for genomic innovation by integrating new genetic material at the *attI* site with minimal disruption to existing genes. The integrated integron cassette can be rapidly expressed via the adjacent Pc promoter, exposing it to selective pressures. Within a bacterial population harboring integrons, individual cells can sample different integron cassettes. This sampling process facilitates the emergence of variants expressing advantageous traits. Notably, the integration of integron cassettes is reversible, allowing their excision as free circular DNA elements. This dynamic ability to integrate and excise integron cassettes enables integrons to capture, rearrange, and shed genetic material in response to environmental selection pressures (Gillings Michael 2014). Recent research has significantly broadened our understanding of integron integrase activity beyond its classical role in site-specific recombination at *attI* and *attC* sites. Souque et al. (2023) demonstrated that, under antibiotic selection pressure, the integrase in *Pseudomonas aeruginosa* can mediate off-target recombination events that result in large-scale chromosomal inversions, particularly within plasmids. These structural changes, driven by integron activity, enhanced bacterial fitness by disrupting energetically costly conjugation genes, thereby promoting compensatory plasmid evolution.

Remarkably, such off-target effects occurred even in the absence of extensive cassette reshuffling, underlining the potential of integrons to facilitate adaptive structural variation and shape genome evolution beyond the exchange of resistance genes (Souque et al. 2023).

Building upon this expanded functional perspective, a subsequent study by Loot et al. (2024) identified a new class of recombination targets—non-canonical chromosomal sites termed *attG*. These sequences, structurally and functionally distinct from classical *attC* and *attI* sites, serve as additional targets for integrase-mediated cassette integration. When located near host promoters, these cassettes can be transcribed and later excised, contributing to gene expression variability and local genomic rearrangements. This newly discovered mechanism reveals yet another layer of integron-mediated plasticity and reinforces the view of integrons as dynamic drivers of horizontal gene transfer and bacterial genome evolution on a broader scale (Loot et al. 2024).

The rise and evolution of integrons: unlocking nature's genetic toolkit

As discussed in the previous section, *integrons* are unique genetic elements known for harboring the crucial gene *intI*, which plays a significant role in their functionality and diversity. They are found in a wide range of bacteria, comprising over 15% of genome-sequenced bacterial species. Integrons are prevalent in diverse environments, such as soil, sediments, biofilms, and marine habitats (Abella et al. 2015; Nield et al. 2001; Stokes et al. 2001).

Five classes of mobile integrons (Class 1 to Class 5) have been identified based on the sequence of their integrase genes (*intI*) (Sabbagh et al., 2021). Classes 4 and 5 integrons have received limited research attention and are yet to be identified in Enterobacteriaceae. Class 4 has been observed in *Vibrio* species, linked to resistance against trimethoprim/sulfamethoxazole, and located within an integrative conjugative element. Class 5 was found on the pRSV1 plasmid of *Vibrio salmonicida*, situated within a complex transposon structure (Hochhut et al. 2001; Sørum et al. 1992). Class 1 integrons are prevalent in various Gram-negative bacteria and significantly contribute to the spread of multidrug resistance. Class 2 integrons, embedded within Tn7 transposons, are also detected in several bacterial species and play a key role in the horizontal transfer of antibiotic resistance genes. Class 3 integrons are less commonly encountered and have been reported in a limited number of species, with comparatively lower impact on resistance dissemination than Classes 1 and 2 (Fluit and Schmitz 2004; Yu et al. 2013). The genomic context of integrons, whether integrated into chromosomes or associated with mobile elements, has significant functional and evolutionary implications. When carried by mobile elements such as plasmid, integrons can disseminate across diverse bacterial species, whereas chromosomal integrons often serve as stable platforms for the accumulation of genetic material, contributing to genome diversification and phenotypic variability (Boucher et al. 2011; Gillings et al. 2005).

Integron distribution patterns suggest occasional horizontal transfer between bacteria occupying similar ecological niches, complementing the primarily vertical inheritance observed across taxa. Phylogenetic analyses of sedentary chromosomal integrons (SCIs) have shown general congruence with host 16S rRNA genes; however, exceptional cases, such as those reported by (Rowe-Magnus et al. 2001), have revealed incongruences, indicating rare horizontal gene transfer events over long evolutionary timescales. These findings support the view that integron systems are ancient and have occasionally crossed taxonomic boundaries under specific environmental pressures. Although integrons themselves are not inherently mobile, their movement depends on associated transposases or recombinases, often located within or near integron cassette arrays. These enzymes facilitate the excision of integrons from chromosomal DNA and their integration into new genomic loci (Cambray et al. 2010; Mazel 2006; Rowe-Magnus et al. 2001).

Major forms of integrons

Mobile integrons (MIs)

Genetic components known as mobile integrons (MIs) can be carried by conjugative plasmids and are often physically linked to mobile DNA elements such as transposons, facilitating the transfer of genetic material between bacterial strains, both within and across species (Stalder et al. 2012). MIs typically contain a small number of integron cassettes with diverse origins, indicating that they have been acquired over time across varied genomic contexts. Despite this diversity, their primary role is frequently associated with antibiotic resistance. A total of 177 distinct integron cassettes have been identified in mobile integrons, as curated in the IntegrAll database using a 95% sequence identity threshold. These cassettes include a wide range of antimicrobial resistance genes, while several others remain functionally uncharacterized (Hipólito et al. 2023; Souque et al. 2021). MIs are classified into three major classes based on the nucleotide sequence of the *intI* gene, and these classes confer resistance to different categories of antibiotics. Notably, integrons belonging to classes 1 and 2 have been historically associated with the widespread dissemination of multidrug resistance (Ayala Nuñez et al. 2022; Mazel 2006). Given their genetic diversity and clinical relevance, a phylogenetic tree was constructed using RAxML, classifying 71 bacterial strains based on *intI* gene sequences. The tree reveals evolutionary relationships among the strains and distinguishes distinct bacterial clades within the dataset (Fig. 2).

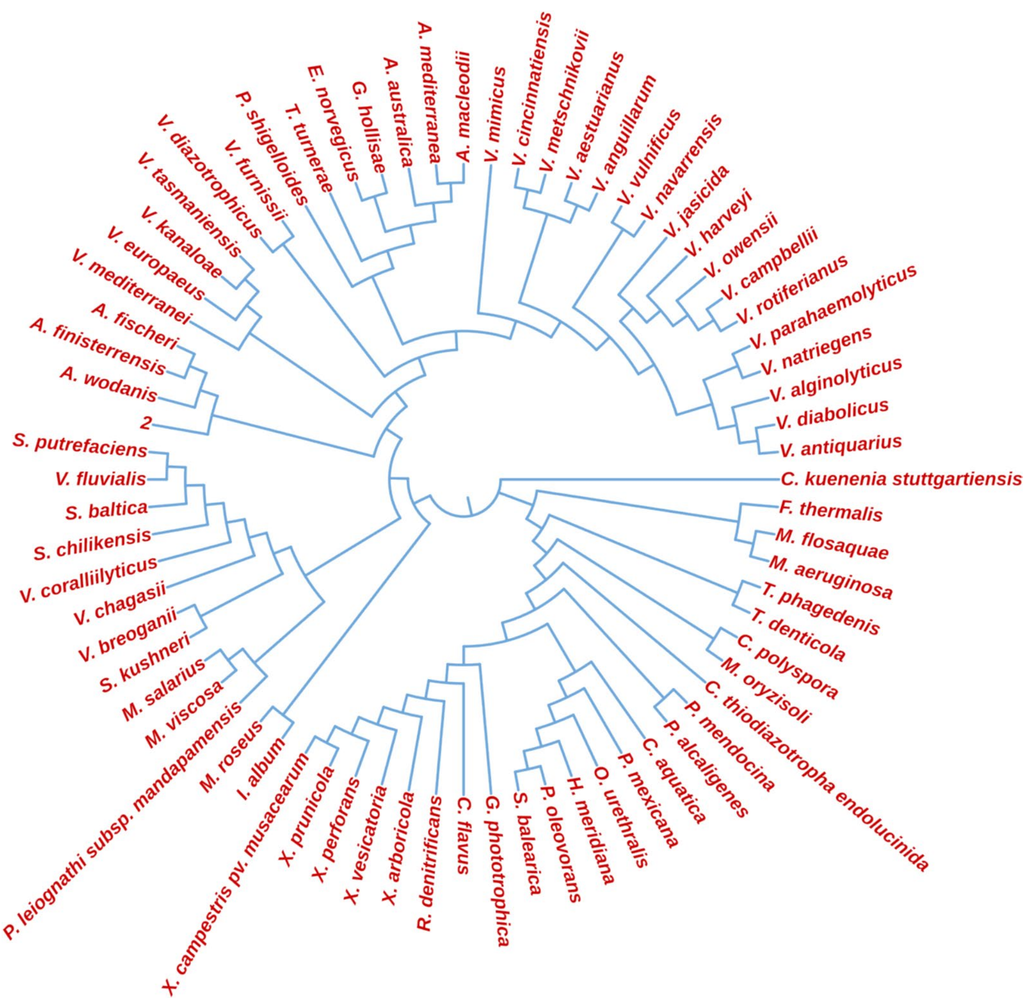


Fig. 2 Circular Hierarchical Cluster analysis (HCA) dendrogram obtained using RAxML (Stamatakis 2014) to verify evolutionary links between different bacterial species based on the "gene sequence of *integrase*" obtained from NCBI. Dendrogram classifying 71 bacterial strains into distinct bacterial groups from variable genus highlighting the relatedness between these bacterial strains signifying their evolutionary relationships based on integrase gene sequences

Class 1 integrons are ubiquitous and clinically significant, commonly found in Gram-negative clinical isolates. They are frequently associated with transposons such as Tn*402* and are often embedded within larger transposons like Tn*21*. Class 1 integrons have been extensively investigated as major contributors to the widespread dissemination of antimicrobial resistance (Sajjad et al. 2011). These genetic elements have been studied in various microorganisms, with their prevalence consistently reported to range from 22 to 59% (Asghari et al. 2021). Notably, class 1 integrons have been identified in clinical Gram-negative bacteria across a wide range of genera, including *Acinetobacter*, *Aeromonas*, *Alcaligenes*, *Burkholderia*, *Campylobacter*, *Citrobacter*, *Enterobacter*, *Escherichia*, *Klebsiella*, *Mycobacterium*, *Proteus*, *Pseudomonas*, *Salmonella*, *Serratia*, *Shigella*, and *Vibrio*. This widespread distribution highlights the critical role of class 1 integrons in the dissemination and persistence of antimicrobial resistance among clinically important pathogens (Yu et al. 2013). Beyond clinical settings, class 1 integrons, particularly the *intI1 gene*—are increasingly detected in diverse environmental niches impacted by human activity. A study have proposed *intI1* as a proxy marker for anthropogenic pollution, as it is commonly associated with resistance to antibiotics, disinfectants, and heavy metals, and is frequently found on mobile genetic elements in wastewater, soil, and aquatic environments (Gillings et al. 2015). This reflects the profound ecological footprint of human-associated microbial resistance genes.

Class 2 integrons are distinct and are exclusively associated with Tn7 transposon variants. They may exhibit variable integron cassette arrays. The integrase gene, *intI2*, typically carries a nonsense mutation that renders it nonfunctional. However, in rare instances, natural suppression mechanisms or recombination activity from other integrases can lead to functional cassette recombination. The 3'-conserved segment (3'-CS) of class 2 integrons contains five genes—*tnsA*, *tnsB*, *tnsC*, *tnsD*, and *tnsE*—which facilitate integron transposition. Unlike class 1 integrons, class 2 integrons contain a nonfunctional *intI2* gene, resulting in a truncated and inactive polypeptide. The class 2 integrase (IntI2) has consistently been found to be non-functional in all sequenced cases due to the presence of an internal stop codon. This is supported by observations of simultaneous carriage of both class 1 and class 2 integrons, limited cassette gene diversity, and distinctive genetic arrangements observed in class 2 integrons. Class 2 integrons are frequently found in certain Gram-negative bacterial genera, including *Acinetobacter*, *Aeromonas*, *Escherichia*, *Morganella*, *Salmonella*, and *Shigella* (Fluit and Schmitz 2004). However, their occurrence and prevalence remain significantly lower than those of class 1 integrons.

Class 3 integrons, although less prevalent than class 2 integrons, are believed to be embedded within transposons. They share a structural organization similar to class 2 integrons. The integrase protein IntI3, identified in soil and freshwater *Proteobacteria*, performs functions comparable to IntI1, including cassette excision and integration. However, despite this functional similarity, differences in recombination efficiency have been observed. IntI3 catalyzes recombination between circularized integron cassettes and the *attI3* site, as well as between 59-base elements (59-be) and secondary sites. However, its activity at secondary sites occurs at significantly lower frequencies than that of IntI1, highlighting functional differences in recombination efficiency between class 3 and class 1 integron integrases. Class 3 integrons were first reported in *Serratia marcescens* and have since been found in a limited number of bacterial species. They are associated with the dissemination of resistance determinants such as the *IMP-1* metallo-β-lactamase gene and occur relatively infrequently in clinical isolates. Reported hosts of class 3 integrons include *Acinetobacter* spp., *Alcaligenes xylosoxidans*, *Citrobacter freundii*, *Delftia* spp., *Escherichia coli*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *Pseudomonas putida*, *Salmonella* spp., and *Serratia marcescens* (Collis Christina et al. 2002; Fluit and Schmitz 2004; Yu et al. 2013).

Another class of integrons, discovered by Mazel et al. (1998), was designated as superintegrons (SIs), with the *Vibrio cholerae* superintegron being the first and most extensively characterized example. This integron, approximately 126 kb in size and comprising nearly 178 gene cassettes, represents the paradigm for Sedentary Chromosomal Integrons (SCIs). Similar SCI structures have been identified in other genera such as *Geobacter*, *Listonella*, *Nitrosomonas*, *Pseudomonas*, *Shewanella*, *Treponema*, *Xanthomonas*, and other marine *Proteobacteria*. These chromosomal integrons are distinct from mobile integrons and are primarily involved in genetic diversification and environmental adaptation, although some cassettes have been associated with resistance to antibiotics such as chloramphenicol and fosfomycin (Akrami et al. 2019; Fluit and Schmitz 2004). Class 5 integrons, though not extensively studied, have not yet been identified in Enterobacteriaceae. However, an important finding has been the detection of a class 5 integron located within a complex transposon on the *pRSV1* plasmid of *Vibrio salmonicida* (Hochhut et al. 2001; Kaushik et al. 2018).

Sedentary chromosomal Integrons (SCIs)

Sedentary Chromosomal integrons (SCIs) are integrated into the bacterial chromosome and are relatively stable compared to mobile integrons. They are typically restricted to specific bacterial lineages and species (Wu et al. 2013). SCIs can carry integron cassettes encoding a diverse array of functions, including antibiotic resistance, metabolic pathways, and virulence factors (Stalder et al. 2012). Unlike mobile integrons, SCIs exhibit limited mobility between bacterial hosts. Nevertheless, they play a vital role in shaping the adaptability and long-term evolution of bacterial genomes by enabling the capture and expression of functionally beneficial genes within a lineage-specific context. SCIs are frequently found in environmental bacteria and, to date, have been comparatively less studied than mobile integrons (Gillings Michael 2014).

SCIs: reservoirs of adaptive functions

Sedentary Chromosomal integron (SCI) arrays play a crucial role in promoting genomic diversity. Even among closely related strains of the same bacterial species, distinct integron cassette arrays are frequently observed. This phenomenon is particularly evident in *Treponema* and *Pseudomonas*, where shared integron cassettes between isolates are relatively rare (Wu et al. 2013, 2012). Studies conducted in *Vibrio cholerae* have revealed frequent gain, loss, and rearrangement of integron cassettes, leading to substantial variation across different strains, serotypes, and even species. The horizontal exchange of integron cassettes between bacteria is widespread, and rearrangements often involve entire blocks of cassettes, allowing the mobilization of multiple linked integron cassettes in a single event. This process contributes to the diversification of cassette content and structural arrangement. In *V. cholerae*, the integron cassette array exhibits remarkable flexibility in generating diversity through repeated rearrangements and cross-species gene exchange (Marin and Vicente 2013). This dynamic recombinational activity facilitates the generation of diverse genotypes that are subject to natural selection, enabling rapid adaptation to changing environmental conditions.

A study on various isolated species of *Vibrio* from coral mucus revealed that only a small fraction of integron cassettes within their integron arrays are shared. Even when identical integron cassettes are present, their positions within the arrays differ, highlighting the rapid evolution of integron architectures in certain *Vibrio* species, potentially at a rate exceeding that observed in *Vibrio cholerae* (Koenig et al. 2011). The activity of the *intI* gene, which mediates rearrangements in integron cassettes, is regulated by environmental and physiological stimuli. Binding sites for LexA, a well-characterized transcriptional repressor controlling the SOS response, are present within the regulatory region of the *intI* gene. Activation of the SOS response induces expression of the integrase protein IntI, significantly increasing the rate of cassette excision. This regulatory mechanism is conserved in both chromosomal and mobile integrons, suggesting an ancestral origin. The SOS response can be triggered by several factors, including foreign DNA uptake, bacterial conjugation, antibiotic exposure, and other stress-related stimuli (Aertsen and Michiels 2006; Erill et al. 2007). Recombination facilitated by integrase also heightened in stationary phase, when the attainment of new functions and genetic diversity can provide significant advantages (Gillings Michael 2014)(Shearer and Summers 2009).

Consistent with these findings, recent research has shown that, SOS regulation enhances cassette rearrangement and gene capture during stress, while effectively ‘freezing’ integron dynamics in stable environments. This regulatory switch is mediated by LexA repression of integrase expression and has been shown to be ancestral and widely conserved across bacterial species. A systematic analysis of integron integrase promoter sequences revealed the prevalence of LexA-binding motifs, highlighting the evolutionary significance of this mechanism in maintaining genomic stability outside of stress conditions

(Cambray et al. 2011). In *Xanthomonas*, the *intI* gene is frequently rendered nonfunctional due to mutations and deletions. In stable ecological niches, continuous activity of the integrase protein IntI may impose a fitness cost on the host organism. Computational modeling provides insights into the selective pressures acting on *intI* genes, suggesting that sustained environmental disturbances may be necessary to preserve functional integrase genes within bacterial genomes. These observations may help explain the widespread occurrence of inactivated *intI* genes and the frequent loss and reacquisition of this gene across bacterial lineages. Together, these features illustrate the unique evolutionary flexibility of integrons in disseminating both gene content and structural diversity under the influence of natural selection (Gillings Michael 2014; Nemergut et al. 2008; Starikova et al. 2012).

Integron cassette and their expression

Expression of integron cassettes in integrons

The expression of integron cassettes within integrons is largely dependent on Pc, the primary cassette promoter (Tansirichaiya et al. 2019). In class 1 integrons, cassette expression is primarily regulated by two Pc promoter variants, Pc1 and Pc2. The strength of the Pc promoter influences the activity of the integrase protein IntI, with weaker promoter variants often associated with enhanced cassette excision activity. The position of an integron cassette within the array also affects its expression. Cassettes located farther from the Pc promoter tend to exhibit reduced transcriptional output. While it has traditionally been believed that cassette expression levels decrease with increasing distance from the Pc promoter, recent findings suggest that this gradient is not solely determined by position. Carvalho et al. (2024) demonstrated that the translation rate of upstream cassettes plays a key role in stabilizing or destabilizing polycistronic mRNA, thereby modulating the expression of downstream cassettes. This translational buffering effect may override promoter proximity, adding a new layer of complexity to integron-mediated gene regulation. Furthermore, they confirm that translational efficiency predominantly governs cassette expression. Their study, which examined expression profiles of 136 antibiotic resistance cassettes within standardized integron platforms, showed that the position of a cassette within the array has only a negligible effect on expression. Instead, they found that the translation rate of upstream cassettes plays a dominant role by influencing the stability of the polycistronic mRNA. Cassettes with reduced translational efficiency were shown to decrease the expression and resistance phenotype of downstream cassettes by destabilizing the entire transcript. These findings propose a revised model of integron expression where translation efficiency, rather than physical distance from the Pc promoter, is the key determinant of cassette expression dynamics (Carvalho et al. 2024).

Furthermore, within integron cassettes, *attC* sites can fold into secondary structures that form stem-loop configurations. These structures may obstruct ribosomal movement during translation, potentially hindering the expression of downstream cassettes within the same array (Jacquier et al. 2009). A recent study by Hipólito et al. (2023) provided one of the most comprehensive experimental profiles of resistance levels conferred by integron-associated cassettes. By cloning 136 antimicrobial resistance cassettes (ARCs) into a standardized class 1 integron expression vector and measuring their minimum inhibitory concentrations (MICs) against eight antibiotic families, the study quantified expression variability across diverse cassettes. Interestingly, even closely related cassettes showed substantial differences in conferred resistance, emphasizing the influence of factors such as translational efficiency, ribosomal access, and genetic context. Moreover, certain *qac* gene variants failed to confer expected resistance levels, challenging long-held assumptions about their functional roles. These findings underscore the complexity of cassette expression and the necessity of considering both sequence and experimental data when assessing resistance potential (Hipólito et al. 2023).

Class 2 integrons and certain sedentary chromosomal integrons (SCIs) also harbor cassette promoters, often located within the *attI* or *intI* regions. In extended cassette arrays, it is likely that internal promoters exist to support expression of specific cassettes. Documented examples of cassettes regulated by internal promoters include those encoding resistance to chloramphenicol and quinolones, toxin-antitoxin systems, and even open reading frameless (ORF-less) cassettes identified in metagenomic datasets (Tansirichaiya et al. 2019). These internal promoters facilitate the expression of individual integron cassettes, regardless of their physical position within the array. A recent study has further expanded the functional repertoire of integron cassettes by demonstrating promoter activity in gene-less, or ORF-less, cassettes from superintegrons in *Vibrio* species. Among 29 gene-less cassettes analyzed, the majority exhibited promoter activity on the sense strand, enhancing the expression of downstream cassettes such as *dfrB9*, which increased trimethoprim resistance over 500-fold in both *Vibrio cholerae* and *Escherichia coli*. Remarkably, one cassette even acted through an antisense promoter, reducing resistance when cloned downstream. These findings highlight the regulatory significance of promoter cassettes within superintegrons and suggest that even cassettes lacking identifiable open reading frames may influence the adaptive potential of integron arrays, particularly if captured by mobile elements (Blanco et al. 2024).

Functional diversity of encoded cassettes

Integrons predominantly carry antibiotic resistance integron cassettes, which provide protection against multiple classes of antibiotics effective against a broad spectrum of bacterial species (Deng et al. 2015; Nemergut et al. 2008). These include aminoglycosides (e.g., streptomycin, neomycin), β-lactams (e.g., penicillin, amoxicillin), chloramphenicol, erythromycin, fosfomycin, lincomycin, quinolones (e.g., ciprofloxacin, levofloxacin), rifampin, streptothricin, trimethoprim, and certain antiseptics (Hall 2013). Unlike mobile integrons (MIs), sedentary chromosomal integrons (SCIs) contain a broader diversity of integron cassettes, many of which have unknown or uncharacterized functions. A significant portion of these cassettes encodes hypothetical proteins or genes with undefined biological roles, highlighting the function of SCIs as reservoirs for genetic innovation and variability (Ghaly et al. 2019). Other cassettes encode functional proteins including phage-associated proteins, toxin-antitoxin systems, acetyltransferases, DNA-modifying enzymes, and virulence factors (Elbourne Liam and Hall Ruth 2006). Metagenomic studies of marine environments have revealed novel integron cassettes encoding proteins involved in iron scavenging (Koenig et al. 2008) and additional acetyltransferases (Levings Renee et al. 2005). Clinically relevant resistance genes have also been identified within cassette arrays. For example, *sul* and *dfr* genes, commonly found in environmental isolates of *Pseudomonas*, *Vibrio cholerae*, and *Vibrio splendidus*, are embedded in integrons (Table 1). When mobilized into MIs, these cassettes can confer high levels of resistance and significantly enhance the adaptability of the host bacterium (Blanco et al. 2024; Rowe-Magnus et al. 2002). Overall, SCIs equip bacteria with a versatile gene pool that includes determinants for antibiotic resistance, virulence, and enzymatic functions, enabling them to respond rapidly to environmental stressors and selective pressures.

Recent advances have uncovered yet another adaptive function encoded by integron cassettes, phage resistance. Escudero et al. (2024) reported that mobile integrons carry bacteriophage resistance integron cassettes (BRiCs), which can be functionally integrated alongside antimicrobial resistance genes. These BRiCs contribute to bacterial immunity against phage attack and show context-dependent fitness effects, modulated by cassette arrangement within the array. Complementing this, a parallel study by Mazel et al. (2024) found that sedentary chromosomal integrons, particularly those from *Vibrio cholerae*, harbor a large number of integron cassettes that encode diverse, and often novel, anti-phage systems. Some of these systems act via cell lysis or growth arrest and include what is now the smallest known protein providing autonomous phage resistance. Together, these findings reinforce the view that integrons act not only as reservoirs of antibiotic resistance but also as versatile defense hubs contributing to microbial survival in phage-rich environments (Darracq et al. 2024; Kieffer et al. 2025). A study revealed that Chilean salmon farms serve as significant reservoirs of bacteria resistant to sulfonamides and trimethoprim, commonly used antibiotics in aquaculture. In particular, bacteria from the *Pseudomonas* genus harboring transferable resistance genes and integrons were highly prevalent, suggesting their role in the environmental dissemination of antibiotic resistance. These findings underscore the urgent need for continuous surveillance to evaluate the potential horizontal transmission of resistance determinants to human pathogens and to better understand fish farming environments as critical reservoirs of antimicrobial resistance (Domínguez et al. 2019).

Table 1 Diverse bacterial species and associated integron cassette arrays in integrons for rapid dissemination of antimicrobial resistance genes

Bacterial species	Class of Integron(s)	Associated resistance GCs/ integron integrases gene	Associated antibiotic Resistance	References
Acinetobacter baumannii	Class 1 & 2	*aadB*	Tobramycin; gentamicin	Anderson Sarah et al. (2018)
		bla$_{OXA-207}$; *bla*$_{OXA-23}$	Carbapenem	Iovleva et al. (2022)
		aphA6; aacA4	Amikacin	Swe-Han and Pillay (2020)
		aacA4-catB8-aadA1	Gentamicin; amikacin	Lin et al. (2010)
	Class 1	*aac(6')-Im (aacA16)*	Amikacin; netilmicin; tobramycin	Harmer Christopher et al. (2023)
		aadA2 and dfrA12	Imipenem; piperacillin; gentamycin; meropenem; ceftazidime; amikacin; trimethoprim-sulfamethoxazole	Khoramrooz et al. (2021)
	Class 1 & 2	*sul1, sul2, tet(B), blaADC, blaOXA, blaTEM, blaCARB, aac(6')-Ib3, aadA1, ant(2'')-Ia, aph(3'')-Ib, aph(3')-Ia, aph(3')-IIa, aph(3')*-VIa, *aph(6)-Id, armA, mph(E), msr(E), catB8*	Amikacin; cefotaxime; ceftazidime; ceftriaxone; cefepime; ciprofloxacin; gentamicin; imipenem; meropenem; trimethoprim/sulfamethoxazole; tetracycline; cefoperazone/sulbactam; piperacillin/tazobactam; colistin; tigecycline; carbapenem	Leungtongkam et al. (2020)
	–	*adeA and adeS gene*	Amikacin; tetracycline; ciprofloxacin; gentamicin	Basatian-Tashkan et al. (2020)
Acinetobacter bereziniae	Class 1	*bla* $_{OXA-23}$	Carbapenem	Abouelfetouh et al. (2022)
		bla $_{OXA-257}$		Zander et al. (2014)
Acinetobacter nosocomialis		*bla* $_{OXA-51}$		Wong et al. (2019)
Acinetobacter radioresistens		*bla* $_{OXA-23}$		Higgins et al. (2013)
Escherichia coli & *Enterobacter* spp.	Class 1	*bla*$_{TEM}$; *bla*$_{TEM-1}$; *bla*$_{SHV-1}$;*bla*$_{CTX-M}$; *aac(3)-IIIa; aac(6')-II; aac(6')-Ie-aph(2''); qnrS; sul1; sul2; sul3; intI1*	Fluoroquinolones; penicillins; carbapenem; sulphonamide	Farkas et al. (2019)

Table 1 (continued)

Bacterial species	Class of Integron(s)	Associated resistance GCs/ integron integrases gene	Associated antibiotic Resistance	References
Klebsiella pneumoniae	Class 1 to 3	*dfrA5; dfrA25; dfrA7; dfrA1-orfC; aadB-catblaOXA10-aadA1; dfrA17-aadA5*	Amoxicillin; cephalothin; nitrofurantoin; ceftazidime; trimethoprim-sulfamethoxazole; tetracycline; gentamycin; nalidixic acid; chloramphenicol; ciprofloxacin; amikacin	Jahanbin et al. (2020)
	Class 1	*qnrA, qnrB, and qnrS*	Meropenem; trimethoprim; ciprofloxacin; nalidixic acid; cefoxitin	Arabzadeh et al. (2022)
	Class 1	*vanTC-02; aacC; blaCTX-M-04; blaSHV-01; blaSHV-02; ampC-04; blaOXY; tetD; blaTEM; tetA-02*	β-Lactamase; aminoglycosides; vancomycin; tetracycline; sulfonamide	Yan et al. (2022)
	Class 1 to 3	*dfrA5; dfrA30; aadA2; aadA2; dfrA12; dfrA17; aadA5; aadA4; dfrA5; dfrA30; aadA2; dfrA5; dfrA30; aadA2; dfrA12; dfrA5; dfrA30; dfrA17; aadA5; aadA4; aadA2; aadA2; dfrA12; dfrA5; dfrA30; aadA2; aadA2; dfrA12; aadA1; dfrA1-sat1; aadA1; dfrA1-sat1*	Ampicillin; Amoxicillin/clavulanic acid; Aztreonam; Cephalothin; Cefotaxime; Ceftazidime; Cefoxitin; Cefteriaxon; Imipenem; Gentamicin; Nalidixic acid; Ciprofloxacin	Firoozeh et al. (2019)
Legionella pneumophila	Class 1	*LpeAB*	Macrolides, fluoroquinolones, rifampicin, doxycycline,	Cocuzza et al. (2021)
			Ciprofloxacin; levofloxacin; moxifloxacin; rifampicin; cefotaxime; tetracycline; trimethoprim/sulfamethoxazole; azithromycin	Natås et al. (2019)
			Macrolide	Massip et al. (2017)
			Rifampin; erythromycin; clarithromycin; azithromycin; ciprofloxacin; moxifloxacin; levofloxacin; doxycycline	Yang et al. (2022)
	–	*lpg1681*	HGA (antimicrobial metabolite)	Levin et al. (2019)
Pseudomonas aeruginosa	Class 1	*aacA4-blaOXA101-aadA5; aacA4; blaOXA-1; blaOXA-101*	Ciprofloxacin; imipenem; meropenem; levofloxacin; β-lactamases; fluoroquinolones; carbapenems	Liu et al. (2020)
	Class 1 to 3	*intI1, intI2 and intI3*	Gentamicin; imipenem; ciprofloxacin; piperacillin; ceftazidime; amikacin; tobramycin	Zarei-Yazdeli et al. (2018)

Table 1 (continued)

Bacterial species	Class of Integron(s)	Associated resistance GCs/ integron integrases gene	Associated antibiotic Resistance	References
Pseudomonas baetica	Class 1	*aac(6')3; qacH; blaOXA-2*	Multiple antibiotics	Vásquez-Ponce et al. (2022)
Aeromonas salmonicida		*dfrA14*	Not specified	
Aeromonas sp.		*blaOXA-956*	Beta-lactam antibiotics	
Salmonella spp.	Class 1	aadA2; aadA5; dfr17	Spectinomycin; streptomycin; trimethoprim	Zhang et al. (2004)
		aadA2; aadA1; drfA1-aadA1; drfA12-aadA2; drfA17-aadA5	Tetracycline; ampicillin	Zhao et al. (2017)
		blaTEM; blaSHV; aadA2; tetA; tetB; sull; dfrA; floR; mphA	β-lactams; aminoglycosides (streptomycin); tetracycline; sulfamethoxazole; trimethoprim; chloramphenicol; macrolides	Abdel Aziz et al. (2018)
Salmonella enteric	Class 1	sul2; strAB; tetAR; aac3-iid; aph; sph; cmy-2; floR; tetB; aadA; aac3-VIa; sul1	Aminoglycosides; β-lactams; tetracyclines; sulfonamides; phenicols; trimethoprim; macrolides; fosfomycin; rifampicin	McMillan et al. (2019)
Staphylococcus aureus	Class 1 & 2	aadB; aadA2; dhfrA1-sat2-aadA1; oxa2; aacA4; orfD-aacA4-catB8; catB3; orfD-aacA4; aadA1; cmlA6; dhfrA11; dhfrA1-sat2	β-lactams; aminoglycosides; streptothricin; trimethoprim; chloramphenicol	Mostafa et al. (2015)
	Class 1	intI1; dhfrV; dfrA1; dfrA12; aadA1; aadA5; aadA4; aadA24; aacA4; aadA2; aadB; cmlA6; qacH; orfX2	Trimethoprim; aminoglycosides; chloramphenicol; quaternary ammonium compound	Li and Zhao (2018)
	Class 1	nuc; mec	Tetracycline; penicillin; clindamycin; erythromycin; linezolid; chloramphenicol; gentamicin; trimethoprim-sulfamethoxazole; oxacillin	Rao et al. (2022)
Vibrio cholerae	Class 1	gene-VCR	Not defined	Mazel et al. (1998)
	Class 1	qacEΔ1 and sul1	Multidrug	Zuberi and Sillo (2022)

Extending beyond classical integrons, emerging resistance genes on mobile elements further enrich the microbial resistome. A novel trimethoprim resistance gene, *dfrA35*, was recently identified within a complex resistance island on broad-host-range IncC plasmids isolated from clinical strains of *Escherichia coli* and *Klebsiella pneumoniae* in Australia. Unlike classical integron gene cassettes, *dfrA35* is not integrated via *attC* sites and does not reside within a known integron structure. Instead, it is located adjacent to a partial CR1 element and surrounded by mobile genetic elements within a multidrug resistance region. Functional validation confirmed that *dfrA35* encodes a trimethoprim-insensitive dihydrofolate reductase, conferring high-level resistance (MIC > 128 µg/mL) when expressed in *E. coli*. Its presence in diverse plasmids and bacterial hosts highlights its potential for horizontal dissemination and clinical significance in antimicrobial resistance (Ambrose and Hall 2019).

Integrons: shaping the landscape of antibiotic resistance

Integrons play a critical role in the rapid dissemination of antibiotic resistance, particularly among Gram-negative bacterial species. They are closely associated with mobile genetic elements such as transposons and conjugative plasmids, which facilitate their efficient transfer both within and across bacterial species. As previously discussed, three major classes of mobile integrons—classes 1, 2, and 3—are commonly encountered in clinical isolates, while two additional classes, class 4 and class 5, are associated with *Vibrio cholerae* and *Alivibrio salmonicida*, respectively. These integrons harbor a diverse array of integron cassettes, many of which encode antibiotic resistance determinants. The extensive repertoire of approximately 130 distinct resistance integron cassettes suggests they have been acquired from diverse sources over evolutionary time (Partridge et al. 2009). Mobile integrons typically feature short cassette arrays regulated by a single promoter, which can limit the expression of downstream cassettes. Nevertheless, these arrays enable resistance to a wide variety of antibiotics used in medical and agricultural contexts (Ali et al. 2020; Ghaly et al. 2021). It is important to emphasize that the shared features observed among antibiotic resistance integrons have emerged through convergent evolution, driven by the intense selective pressure resulting from the widespread use of antibiotics in human activity (Ghaly et al. 2021).

Comparing clinical integrons: Class 1, Class 2, and Class 3

To facilitate a comprehensive understanding of the distinguishing characteristics of clinically significant integrons, a comparative overview of Class 1, Class 2, and Class 3 integrons is presented in Table 2. Although these integrons share a conserved structural framework and a common function in capturing and expressing integron cassettes, they differ markedly in terms of integrase gene functionality, association with mobile genetic elements, prevalence in clinical isolates, and the repertoire of resistance determinants they harbor. This comparative analysis highlights their respective roles in the horizontal dissemination of antimicrobial resistance and their broader implications for bacterial adaptation and genome evolution.

Recent advances: "IntegronFinder": a valuable tool for integron detection and analysis

In the current era of rising antimicrobial resistance, accurate detection and analysis of integrons has become increasingly crucial for understanding their evolutionary dynamics and mitigating the spread of resistance. To address this need, IntegronFinder has emerged as a state-of-the-art computational tool designed to efficiently identify and characterize integrons in microbial genomes. Integronfinder's workflow involves a multi-step approach that integrates sequence similarity searches, conserved domain identification, and integron cassette clustering algorithms. It leverages well-established bioinformatics tools such as BLAST, HMMER, and CD-HIT to ensure robust and reliable detection. Moreover, the implementation of customizable parameters allows researchers to tailor the analysis to specific research objectives or genome contexts. IntegronFinder generates comprehensive reports containing information on (i) integron boundaries, (ii) integron cassette content, and (iii) associated functional domains. In addition to textual output, the tool provides intuitive visual

Table 2 Comparative Overview of Clinically Relevant Integrons

Feature	Class 1	Class 2	Class 3	References
Integrase Gene	*intI1* (functional)	*intI2* (contains an internal stop codon)	*intI3* (functional)	Arakawa et al. (2000); Hansson et al. (2002); Recchia et al. (1994)
IntI Recombination Sites	Recombines *attI1* × *59-be*, *59-be* × *59-be*, and *attI1* × *attI1*; highest efficiency with *attI1* × *59-be*; full *attI1* required for optimal activity	Recombination region spans from *attI2* to ~ 316 bp downstream; relatively conserved among class 2 integrons	Recognizes and recombines *attI3* and cassette-associated *59-be* sites; catalyzes both integration and excision of gene cassettes	Collis Christina et al. (2002); Jové et al. (2017); Partridge et al. (2000)
Regulatory Mechanism of Integron Recombination	SOS-response via LexA binding sites	Not SOS-regulated; two active Pc promoters in *attI2*; most integrases inactive; active variants linked to weaker promoter activity	Primarily through a promoter, Pc3, located near the intI3 gene	Baharoglu and Mazel (2011); Fonseca and Vicente (2022); Guerin et al. (2009); Jové et al. (2017); Lacotte et al. (2017)
Associated Transposons	Tn*402*-like transposons and the Tn*3*-family (e.g., Tn*21*, Tn*1696*)	Commonly associated with Tn7-family transposons, including Tn7, Tn*1825*, Tn*1826*, and Tn*4132*	Tn*402* (reverse orientation)	Deng et al. (2015); Kaushik et al. (2018); Xu et al. (2009)
Mobility	Ancestrally non-mobile; association with Tn*402* carrying the complete *tniABQC* module and flanked by 25 bp IRi and IRt enables self-mobility	The 3′-CS of the Tn7 transposon encodes *tnsA–E* genes, which mediate transposon mobility via site-specific insertion into bacterial chromosomes	Less mobile; mobility less characterized	Domingues et al. (2012); Gillings Michael (2014); Hansson et al. (2002); Labbate et al. (2009); Partridge et al. (2009); Senda et al. (1996)
Cassette Array Conservation	Highly variable	More conserved compared to class 1 integron	Limited characterization	da Fonseca et al. (2011); Fluit and Schmitz (2004)
Functional Promoters	Pc promoter drives gene cassette expression; ≥ 13 variants described. Four major variants—PcW, PcH1, PcWTGN-10, and PcS—differ in strength. Promoter strength inversely correlates with integrase activity; location within *intI1* affects both Pc function and IntI1 sequence	Weak or absent Pc; recombination activity low	Pc promoter is present	Collis Christina et al. (2002); Hall and Collis (1995); Jové et al. (2010)

Table 2 (continued)

Feature	Class 1	Class 2	Class 3	References
Host Range and Common Gene Cassettes	Extensively distributed among Gram-negative pathogens (*Escherichia, Pseudomonas, Salmonella, Vibrio, Acinetobacter, Klebsiella*) and reported in Gram-positive genera (*Corynebacterium, Streptococcus, Enterococcus, Staphylococcus, Aerococcus, Brevibacterium*). Frequently detected gene cassettes include *aadA*, *dfrA*, and *bla* (β-lactam resistance)	Primarily found in Gram-negative bacteria including *Acinetobacter*, *Salmonella*, *Pseudomonas*, *Enterobacteriaceae*; occasionally detected in *E. coli*, *Proteus*, *P. aeruginosa*, and *E. faecalis.* Common cassettes: *dfrA1*, *sat1*, *aadA1*; rare variants include *ereA*, *aadB*, *catB2,* and novel rearrangements	Primarily reported in low frequencies across *Serratia marcescens*, *Klebsiella pneumoniae*, *Escherichia coli*, *Acinetobacter spp.*, *Citrobacter freundii*, *Pseudomonas spp.*, *Salmonella spp.*, and *Alcaligenes*. Commonly associated with *bla _{GES-1}* and *IMP-1* metallo-β-lactamase gene cassettes	Deng et al. (2015)
Clinical Impact	High; major contributor to multidrug resistance	Exhibit lower prevalence compared to class 1	Emerging; low prevalence in clinical settings	Deng et al. (2015); Gillings (2017); Kaushik et al. (2018)
Resistance Genes / Associated Antibiotics	*dfrA* (trimethoprim resistance), *sul1* (sulfonamide resistance); confers resistance to co-trimoxazole (TMP-SMX) by interfering with folic acid synthesis enzymes	*dfrA1* (trimethoprim resistance), *sat1*, *sat2* (streptothricin resistance), *aadA1*, *aadB* (streptomycin/spectinomycin resistance), *catB2* (chloramphenicol resistance), *ereA* (erythromycin resistance)	*blaGES-1* (β-lactam resistance), *IMP-1* (metallo-β-lactamase; broad-spectrum β-lactam resistance)	Arakawa et al. (1995); Biskri and Mazel (2003); Collis Christina et al. (2002); de los Santos et al. (2021); Hansson et al. (2002); Ramírez María et al. (2005); Xu et al. (2009)

representations, such as graphical maps and network diagrams, that facilitate the interpretation of integron structures and cassette arrangements.

Several important investigations have employed IntegronFinder as a powerful computational tool to unravel the complexity of integrons. Notably, the study highlighted the pivotal role of IntegronFinder in advancing antimicrobial resistance (AMR) research. The researchers used the software to identify group II introns associated with integrons and CALINs (Clusters of *attC* sites Lacking Integron integrase) in halophilic genomes and metagenomes derived from hypersaline environments (Fig. 3). Their findings provided novel insights into the diversity and distribution of group II introns in halophiles, emphasizing their potential biotechnological relevance in microbial evolution. IntegronFinder was instrumental in accurately identifying and characterizing these introns across the studied datasets (Sonbol and Siam 2021).

Another study by demonstrated the capabilities of IntegronFinder version 2, wherein they analyzed over 20,000 fully sequenced bacterial genomes to explore the global distribution of integrons. Additionally, they focused on approximately 4,000 genomes of the pathogenic bacterium *Klebsiella pneumoniae*, assessing integron content and its link to antibiotic resistance.

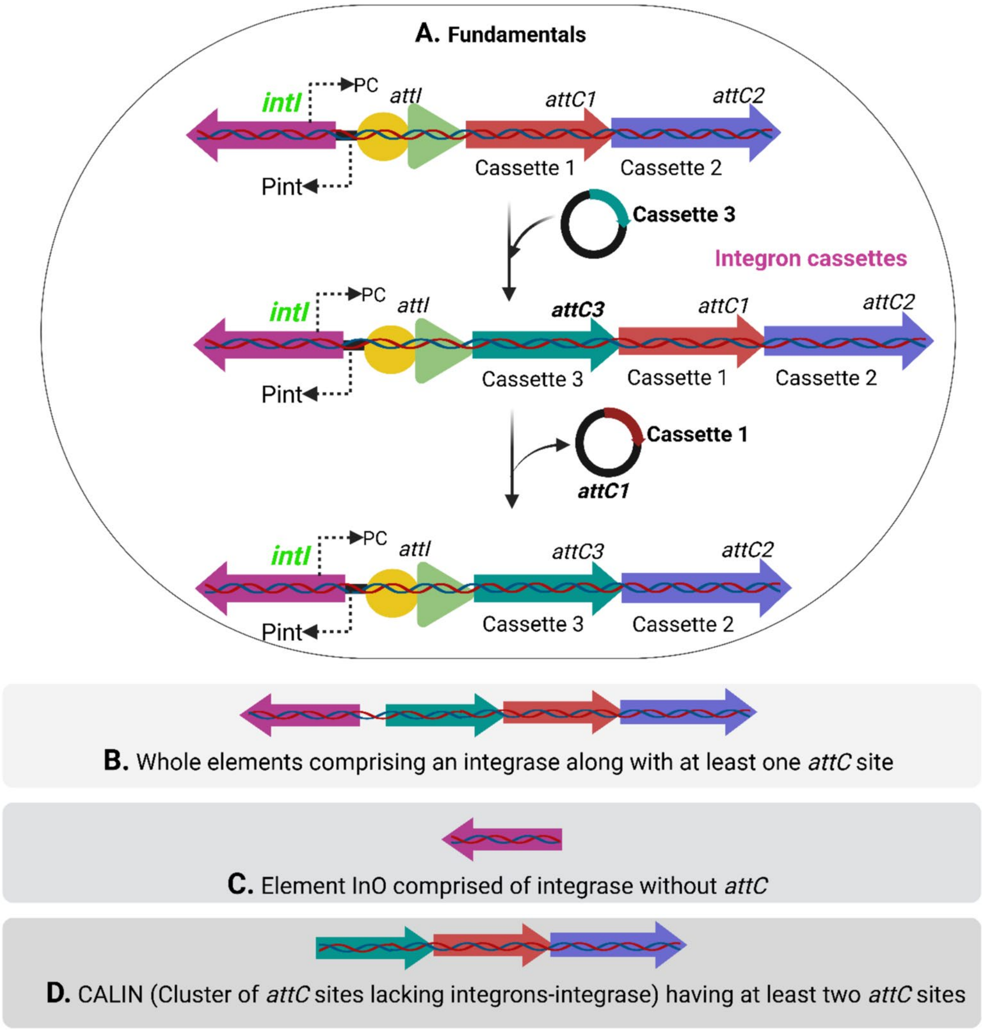


Fig. 3 Illustration represents **a** fundamental structure of integrons and integration of integron cassettes; **b**, **c** and **d** different elements of integrons subjected for IntegronFinder based analysis

The study revealed that *K. pneumoniae* harbors a wide variety of integrons, including the largest mobile integron identified in the plasmid database. Furthermore, the integrons' pangenome was found to include around 165 distinct gene families, primarily associated with antibiotic resistance mechanisms (Néron et al. 2022). These studies highlight the significance of IntegronFinder as a free and open-source software tool that facilitates comprehensive integron detection and annotation. Its application continues to expand our understanding of genomic regions associated with antimicrobial resistance and their evolutionary trajectories.

A comprehensive overview detailed as Fig. 4 showcases a systematic and well-structured outline of the sequential steps necessary for identifying integrons using the powerful IntegronFinder tool. It aims to enhance comprehension and streamline the complex procedure, enabling researchers to seamlessly integrate it into their investigations and analyses.

Conclusions

The field of integron research holds immense potential for deepening our understanding of bacterial evolution and adaptation. Despite notable progress, several critical questions remain unanswered. For example, the mechanisms by which integron cassettes within class 1 integrons and superintegron arrays are expressed, and how they contribute to host fitness, are not yet fully understood. Similarly, the molecular mechanism by which a single recombinase recognizes both single-stranded and double-stranded recombination sites remains unresolved. However, recent work has begun to shed light on these recognition

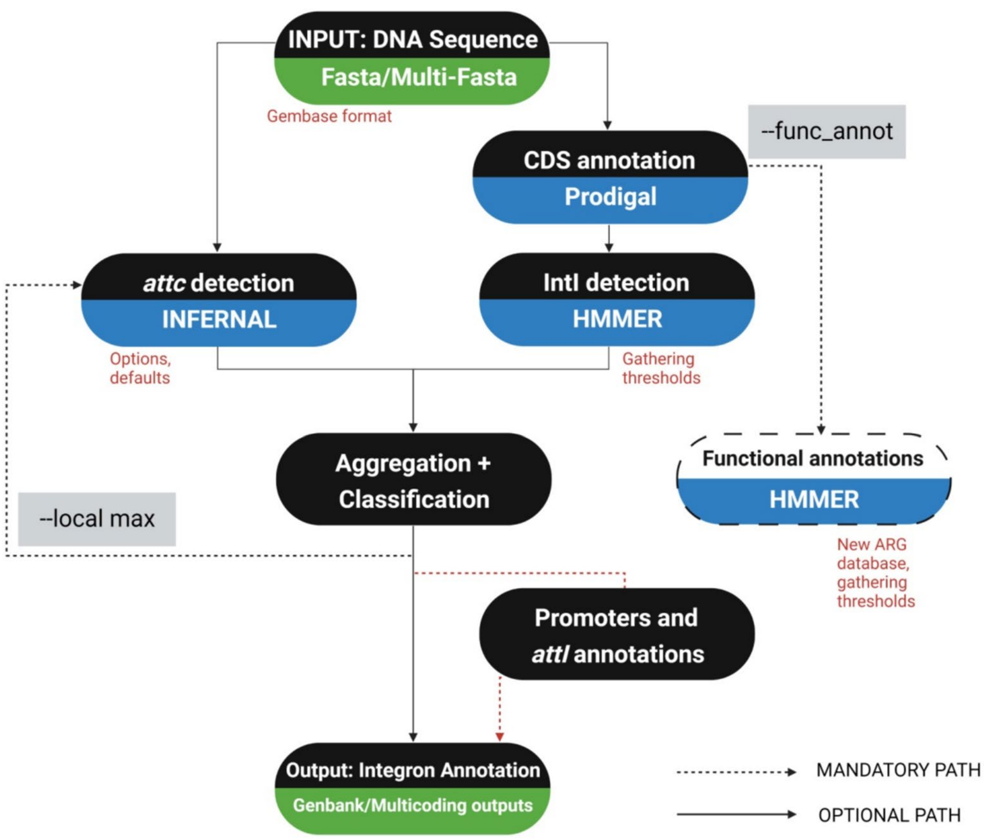


Fig. 4 Steps involved during the computational-exercise associated with IntegronFinder for the identification and annotation of integrons (Cury et al. 2016; Néron et al. 2022)

mechanisms. Complementary studies employing directed evolution of IntI1 further identified key domains that modulate *attI* × *attI* recombination, laying the groundwork for understanding how integrase accommodates both single- and double-stranded recombination substrate. These findings have laid a strong conceptual foundation, yet a complete understanding will require detailed structural and biochemical studies. In particular, elucidating the three-dimensional structure of the integrase–*attI* complex will be critical to fully unravel how integrase accommodates and processes different recombination substrates. Another area warranting attention is the generation of new integron cassettes and the complex relationship between SOS induction and horizontal gene transfer. Investigating these interactions may reveal key insights into the coordinated adaptive responses of bacterial communities. Moreover, exploiting the connection between integron dynamics and the SOS response presents an opportunity to develop innovative strategies for controlling the spread of antibiotic resistance in clinical settings. By exploring these aspects more deeply, researchers can better define the functional roles of integron cassettes within integrons and their evolutionary significance. Such knowledge will enhance our ability to predict bacterial adaptation in environmental and clinical contexts and may inform novel interventions to mitigate antibiotic resistance. Tools like IntegronFinder will continue to play a vital role in this process by enabling the efficient detection and analysis of integrons in diverse microbial genomes. The software provides comprehensive reports on integron boundaries, *GC* content, associated functional domains, and intuitive visualizations. Its effectiveness has been demonstrated in numerous studies that accentuate its value in characterizing integrons and their distribution across bacterial species. Overall, the future of integron research is promising. Continued investigations will undoubtedly lead to transformative discoveries with both theoretical and practical implications in microbial genomics, antibiotic resistance, and evolutionary biology.

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Declarations

Competing Interest The authors declare no competing interests.

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